(Deemed to be University under section 3 of the UGC Act, 1956)

Final Assessment Test(FAT) - Nov/Dec 2024

Programme	**B.Tech.**	Semester	**Fall Semester 2024-25**
Course Code	**BMAT205L**	Faculty Name	**Prof. Rajivganthi**
Course Title	**Discrete Mathematics and Graph Theory**	Slot	**D1+TD1+TDD1**
		Class Nbr	**CH2024250102081**
Time	**3 hours**	Max. Marks	**100**

General Instructions

- Write only Register Number in the Question Paper where space is provided (right-side at the top) & do not write any other details.

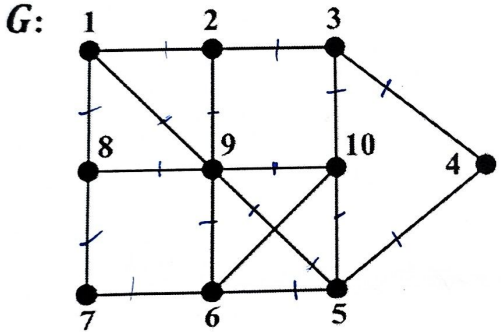
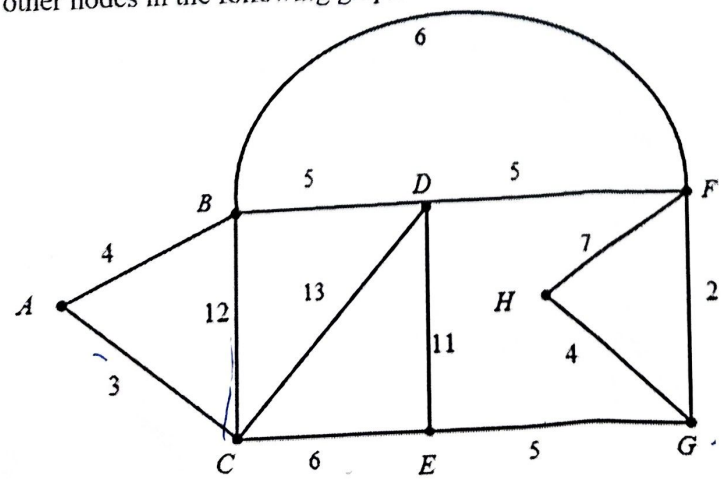
Course Outcomes

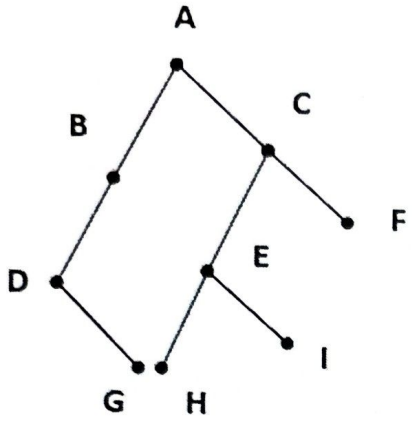
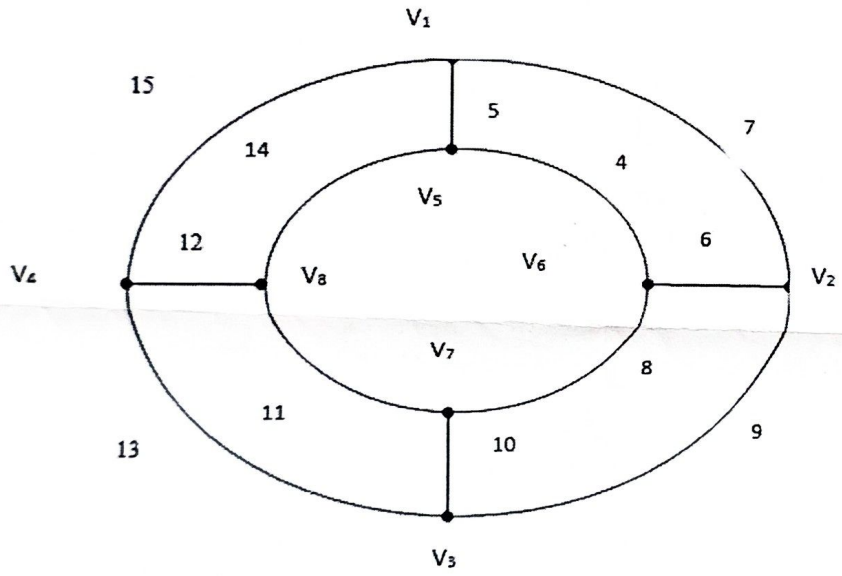
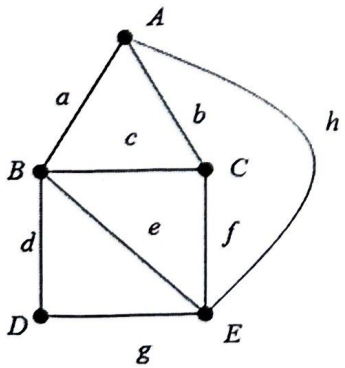
1. Learn proof techniques and concepts of inference theory
2. Use algebraic structures in applications
3. Counting techniques in engineering problems.
4. Use lattice and Boolean algebra properties in Digital circuits.
5. Solve Science and Engineering problems using Graph theory.

Section - I

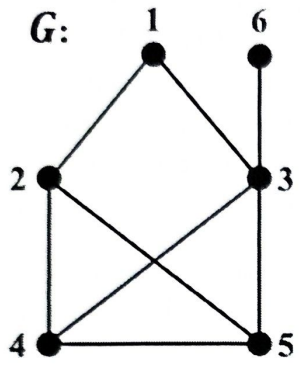
Answer any 10 Questions (10 × 10 Marks) *M - Marks

Q.No	Question	*M	CO	BL
01.	A) Using the rules of inference, construct a valid argument showing that the conclusion "Someone who passed the first exam has not read the book" follows from the premises "A student in this class has not read the book" and "Everyone in this class passed the first exam". Assume that the Universe of Discourse consists of all the people in this world. **[6 Marks]** B) Show that $r \to s$ can be derived from the premises $\neg r \vee p$, $p \to (q \to s)$ and q. **[4 Marks]**	10	1	3
02.	A) Prove the logical equivalence $((p \vee q) \wedge (p \vee \neg q)) \vee q \equiv p \vee q$. **[4 Marks]** B) Find the PDNF and PCNF of $(\neg p \to r) \wedge (q \leftrightarrow p)$. **[6 Marks]**	10	1	4
03.	Let $(A, \triangle)$ and $(B, *)$ be two Abelian groups. An operation $\square$ is defined on the Cartesian product $A \times B$ as follows: For any (a_1, b_1) and (a_2, b_2) in $A \times B$, $(a_1, b_1) \square (a_2, b_2) = (a_1 \triangle a_2, b_1 * b_2)$. Verify whether $(A \times B, \square)$ is an Abelian group?	10	2	5
04.	A) Given is the generator matrix for an encoding function $e : B^3 \to B^6$: $G = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix}$. Decode the following received words: (i) 110010 (ii) 010110 (iii) 100011 (iv) 100100 by constructing parity-check matrix. **[6 marks]** B) (i) How many different ways can 9 people be arranged in a row if: (a) three specific people must always stay together, (b) three specific people must never stay together? **[4 marks]**	10	2,3	3

05.	A) In a school of 200 students, it is observed that; 85 students study Mathematics, 70 students study Science, 60 students study English, 30 students study both Mathematics and Science, 25 students study both Mathematics and English, 20 students study both Science and English, 15 students study all three subjects. How many students study at least one of these subjects, and how many students do not study any of these subjects? **[7 marks]** B) In a group of 30 people, each person has a birthday on one of the 12 months of the year. At least how many people share the same birth month? **[3 marks]**	10	3	5
06.	A) Convert the Boolean expression $x' + x(x + y')(y + z')$ into sum of products and product of sums. **[5 Marks]** B) Use Boolean algebra to simplify the expression $(A + C)(AB + AB') + AC + C$, then obtain a logic gate circuit diagram for the simplified expression. **[5 Marks]**	10	4	5
07.	A) Check whether (S, R) is a POSET where S is the set of all people in the world and R is the relation defined on set S such that $(x, y) \in R$ where x and y are people, if x is not taller that y . **[5 Marks]** B) Find at least two distinct minimal edge coverings and two distinct minimal vertex coverings for the following graph G: **[5 Marks]** **G:** 	10	4	4
08.	i) Draw a simple graph represented by the following incidence matrices where rows and columns denote vertices and edges respectively of the graph G. **[6 Marks]** $A = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 1 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 1 \end{pmatrix}$ ii) Are the graphs obtained from the above matrices isomorphic? Justify. **[4 Marks]**	10	5	4
09.	Apply Dijsktra's algorithm to find the shortest path and shortest distance from the node **A** to all other nodes in the following graph. 	10	5	3

10.	A) Construct the binary tree whose inorder: H D I B J E K A F C G and post order H I D J K E B F G C A. **[5 Marks]** B) Trace the inorder, preoder and postorder tree traversals for the following graph. **[5 Marks]** 	10	5	5
11.	A) Using Prim's algorithm, find the minimum spanning trees of the weighted graph given below, **[7 Marks]**  B) Draw all the fundamental circuits from the following graph **[3 Marks]** 	10	5	4

12.	Find the chromatic number and the chromatic polynomial for the following graph	10	5	5



BL-Bloom's Taxonomy Levels - (1.Remembering, 2.Understanding, 3.Applying, 4.Analysing, 5.Evaluating, 6.Creating)

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